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Cryptocurrency Price Prediction Based Social Network Sentiment Analysis Using LSTM-GRU and FinBERT

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ABSTRACT Cryptocurrencies are digital assets that are widely used for trading and investing. One of the characteristics that traders take advantage of for profit is the high volatility of the price. Its volatile and rapidly changing prices have made cryptocurrency price predictions a challenging and highly sought-after research topic. Cryptocurrency price predictions usually only use historical prices on the dataset, while price movements are also influenced by other aspects such as sentiment contained in social media. This study proposes a new machine learning method to predict Ethereum and Solana cryptocurrency price, which integrates cryptocurrency historical price data and social media sentiment as inputs of the prediction model. FinBERT, a pre-trained sentiment analysis model is used to extract the sentiment implied in social network tweets into daily sentiment score, which are then combined with the historical market price data. The hybrid model of LSTM-GRU model is used to train the dataset and perform cryptocurrency price prediction. The experiment results show that the presented method can successfully predict the Ethereum and Solana price movement and has superior performance than all the benchmark models.

INDEX TERMS FinBERT, Social network, Sentiment Analysis, Hybrid LSTM-GRU, Ethereum Prediction, Solana Pediction.

I. INTRODUCTION

Cryptocurrency is a digital and financial asset that has revolutionized the financial sector with its rapid development, especially in recent years [1]. Bitcoin as a pioneer of cryptocurrencies, can overcome some of the problems faced by traditional economic systems. In the traditional economic system, payments or transactions can only be made with the help of third parties such as banks, so there can be some shortcomings in aspects of trust, transparency, flexibility, and security. With bitcoin, online transactions from both parties can occur directly without intermediaries using blockchain technology [2]. Every transaction is permanently recorded in chain of blocks and new transaction is validated using proof-of-work consensus mechanisms, which can prevent parties from creating fake transactions or trying to manipulate records of transactions that have occurred [3]. The virtual currency that exchanged or transacted using blockchain technology is called cryptocurrency. In the above case, the cryptocurrency is bitcoin itself.

There are many cryptocurrencies that are used in many blockchain projects. Those cryptocurrencies have their own function and usage, but one of the natures of cryptocurrency is its always changing price. Cryptocurrency prices are relatively much more volatile than other traded assets such as stocks, foreign exchange, gold, etc. Due to its price volatility, traders use it to seek profits from the difference between buying and selling prices [4]. The price of cryptocurrency has piqued the interest of many researchers around the world. Researchers have been trying various methods to predict the volatile price of cryptocurrencies, especially in the field of machine learning [5]. Machine learning has been known as one of the excellent methods for prediction. Some machine learning algorithm like XGBoost and LSTM (Long Short

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Term Memory) has exceed the performance of moving average method to predict cryptocurrency price [6]. Performance of various machine learning algorithms have been compared in dealing with cryptocurrency price prediction or forecasting problem [7]–[9]. Recurrent Neural Network (RNN) algorithm such as LSTM and GRU has scored a better cryptocurrency price prediction performance than ARIMA (Autoregressive Integrated Moving Average), one of the machine learning methods commonly used in price forecasting [10]. Hybrid model of LSTM-GRU have the highest performance score than the comparison algorithms in regards to handle foreign exchange price prediction [11].

Cryptocurrency price prediction or forecasting has been a challenging and difficult subject of research due to its unstable and rapidly changing price. Cryptocurrency price volatility is caused by several factors such as popularity, transaction cost and speed, market trends, news, public sentiment and some other factors [12]. Sentiment in social media has an important correlation and influence on cryptocurrency price movements [13]–[15]. One example is a tweet from Elon Musk about one of the famous cryptocurrencies, doge coin. Bitcoin and doge coin price has gone up and down rapidly mainly because of that particular tweet [16]. Many traders make buying and selling decisions based on the macro and micro public sentiment on social media.

Without a doubt, bitcoin is the most famous cryptocurrency and most of the research regarding cryptocurrency price prediction revolves around bitcoin. As mentioned before, cryptocurrency is the currency for a specific blockchain project. There are several aspects of blockchain that make many parties want to build their projects using blockchain technology, one of which is the decentralized nature of blockchain so that no single large party owns or can manage every activity and manipulate the data that recorded on the blockchain [17]. With smart contract feature, users can build their application or program on top of the blockchain network, so the application will be decentralized and protected from hacker attacks [18]. Ethereum is a blockchain network with smart contract features that has the highest number of adoptions, and its cryptocurrency, Ether (ETH) is the cryptocurrency with the second largest market capitalization after Bitcoin [19]. Another widely adopted smart contract blockchain network that offer faster transaction speed than ethereum is Solana, with its token, SOL [20].

A. MOTIVATION

In this research, the daily closing price of ethereum and solana cryptocurrency (ETH and SOL) will be predicted using hybrid RNN machine learning architecture, LSTM-GRU. Recurrent Neural Networks (RNNs) are a type of neural network architecture commonly used for sequential data processing. However, they suffer from a significant weakness known as the "vanishing gradient" problem. The vanishing gradient problem arises during the training phase of RNNs, particularly when using backpropagation through time (BPTT) to update the network's weights. As the network processes sequential data, it calculates gradients that indicate how much each weight should be adjusted to minimize the error in the predictions. These gradients are backpropagated through time to update the network's parameters. The issue occurs when the gradients become extremely small as they are backpropagated from the later time steps back to the initial time steps. This happens because the gradients are calculated through the multiplication of many intermediate gradients through time. As these gradients get multiplied, their values decrease exponentially, causing them to vanish or become very close to zero. LSTM and GRU are RNN architecture designed specifically to address the vanishing gradient and exploding gradient problems. LSTM and GRU hybrid architecture is one of the best machine learning algorithm for time-series forecasting such as cryptocurrency price prediction [21].

Although capable of solving the vanishing gradient and exploding gradient problems, they have different gates. LSTM has three gates: the input gate, the forget gate, and the output gate. GRU has two gates: the reset gate and the update gate. In certain cases, GRU may be simpler and require fewer parameters, while LSTM can have a larger memory capacity but be more computationally complex [11]. Therefore, combining these two algorithms, it is expected to create a dynamic gate which optimizes model.

Moreover, the influence of sentiment analysis is considered to be able to influence stock price movements as seen from several previous studies [22] [23] .

B. CONTRIBUTION

Combining LSTM and GRU is designed to solve the vanishing gradient and exploding gradient problems. It creates a balance between the complexity and memory capacity required by the model. It helps control the flow of information and gradient flow within the network. The attributes used for these cryptocurrency price predictions are daily closing, opening, high, and low price.

This research also takes advantage of the sentiments contained in the social network 's tweets and combined with historical cryptocurrency data such as daily closing, opening, high, and low price to predict the daily closing price of the cryptocurrency in the daily timeframe. Tweet sentiment data is obtained using FinBERT, a model from BERT that focuses on dealing with Natural Language Processing (NLP) problems in the financial context.

Therefore the main contribution of this paper is to build a crypto currency price prediction model based on the LSTM-GRU algorithm and including Social network sentiment data extracted with FinBERT.

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The evaluation of LSTM, GRU, LSTM-GRU are conducted and comparing between combining with sentiment and without sentiment data. The performance is calculated in MSE, RMSE, MAE, and MAPE.

The remainder of this paper is organized as follows. Section II gives a brief related works of this study. Section III provides a detailed description of the proposed method. Section IV presents the analysis and the performance, while conclusions are offered in Section V.

II. RELATED WORKS

A. PREVIOUS RESEARCH

With the rapid developments in the world of cryptocurrency and blockchain technology, as well as the increasing number of cryptocurrency traders and investors, many are conducting research in the field of prediction, forecasting, and sentiment analysis on cryptocurrency prices. Traders often use trading indicators to help them make trading decisions. As for price prediction, machine learning algorithms such as XGBoost and LSTM (Long Short Term Memory) outperform Moving Average, one of the most used trading indicator [6]. ARIMA (Autoregressive Integrated Moving Average), a well-known and widely used machine learning algorithm to perform time-series forecasting, have better performance and error rate than XGBoost in Bitcoin price prediction [24].

As machine learning technology developed, an algorithm that can predict and remember historical data in its process, called Recurrent Neural Network (RNN) began to emerge. RNN algorithm such as LSTM and GRU (Gated Recurrent Unit) has scored a better cryptocurrency price prediction performance than ARIMA in predicting Bitcoin price [10]. Research in Foreign Exchange (Forex) price prediction combined LSTM and GRU into a hybrid model. Turns out, the LSTM-GRU model has the best result in performance compared to other several comparing algorithms [11].

Evaluation result from a research that conducted social media sentiment analysis on cryptocurrency price fluctuations shows that people's behavior on social media, social network and search on Google, has a significant correlation and impact on the price of cryptocurrencies [25]. One of the best algorithms to perform sentiment classification from text data is Naïve Bayes algorithm, as shown in a study that has compared 10 classifier methods. 5 of them using the lexicon approach, and 5 other methods using the machine learning approach [26]. In an experiment on stock market sentiment analysis, algorithm Naïve Bayes algorithm and 4 other algorithms appears to have lower performance when compared to BERT (Bidirectional Encoder Representations from Transformers) [27].

BERT has been used by developers to create many pre-trained machine learning algorithms for specific needs, and one of them is FinBERT. FinBERT is a pre-trained Natural Language Processing model aimed at the financial sector that increases 15% accuracy in state of the art NLP in the financial sector [28]. A study that uses several machine learning algorithms and sentiment analysis in predicting several cryptocurrencies by utilizing market price data and sentiment from social media Social network , resulted in the conclusion that it is very possible to predict cryptocurrency prices using social network sentiment data and historical market prices [29].

B. LONG SHORT TERM MEMORY

Long Short Term Memory, or commonly abbreviated as LSTM, is a neural network architecture derived from the Recurrent Neural Network or RNN. RNN is a neural network algorithm that can remember information from previous inputs. However, it cannot remember information with a period of more than 5-10 discrete time on sequential data or a long period of time, this problem is called the long term dependencies problem [30]. Because of this problem, information that has been lost or forgotten for a long time is called a vanishing gradient. LSTM is here to fix the vanishing gradient problem by adding a memory cell that can learn which information to remember and which to forget [31].

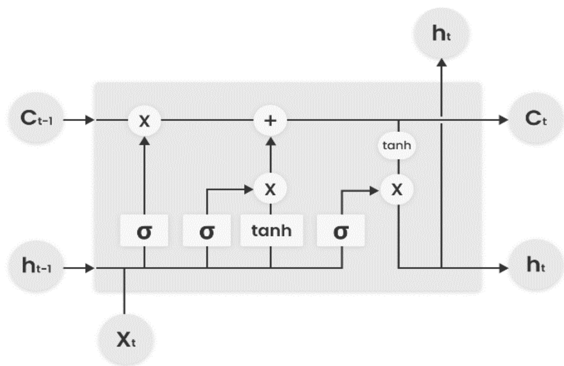

FIGURE 1. **LSTM Architecture**

The main components of an LSTM cell are as follows:

Cell State (*Ct*): The cell state acts as a memory and flows through the entire sequence of the data. It can preserve information over long periods, making it ideal for handling long-term dependencies.

Input Gate (*i*): This gate determines how much of the new information should be added to the cell state.

Forget Gate (*f*): The forget gate decides how much of the previous cell state should be retained and passed along to the next time step. This helps the LSTM forget irrelevant information.

Output Gate (*o*): The output gate controls how much of the cell state should be exposed to the output and passed to the next hidden state.

As shown in Figure 1, C_{t-1} is the cell state of the previous cell, h_{t-1} is the output of the previous cell, C_t is the current cell, h_t is the output current cell, and X_t is the input to the current cell. The information that has been obtained will be

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filtered by the input gate, forget gate, and output gate, to obtain which information should be remembered and which should be forgotten.The operations within an LSTM cell are based on sigmoid and element-wise multiplication operations, which allow the network to learn when to add new information, forget old information, and output relevant information. The LSTM architecture has been widely used in various applications, such as natural language processing, speech recognition, time series prediction, and more. Its ability to handle long-term dependencies and prevent gradient-related issues makes it a powerful choice for sequence modeling tasks.

C. GATED RECURRENT UNIT

Gated recurrent unit or GRU is one of the architectures of the RNN. Like LSTM, GRU also fixes the vanishing gradient problem that occurs in RNNs. The GRU has a similar architecture to the LSTM but is simpler because the GRU does not have a cell state (C_t) and there are fewer gates in the GRU. The GRU has the output of the previous cell, h_{t-1} and the input to the current cell, X_t which generates the output of the current cell. The GRU has a reset gate for short term memory, and an update gate for long term memory. The architecture of the GRU is depicted in Figure 2.

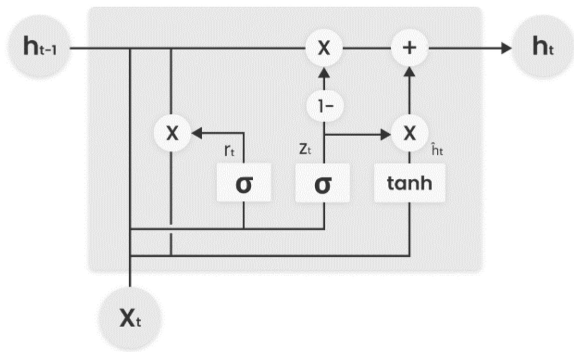

FIGURE 2. GRU Architecture

11 problems in the NLP field [32]. BERT is a modification of Transformer architecture which is designed to pre-train the bidirectional representation of unlabeled text data. The pre-trained BERT model can be modified by adding an output layer to solve various problems that have specific needs and goals (fine tuning).

There are two main stages in BERT, namely pre-training and finetuning. The pre-training stage is a process where BERT is trained to understand the language or text that is

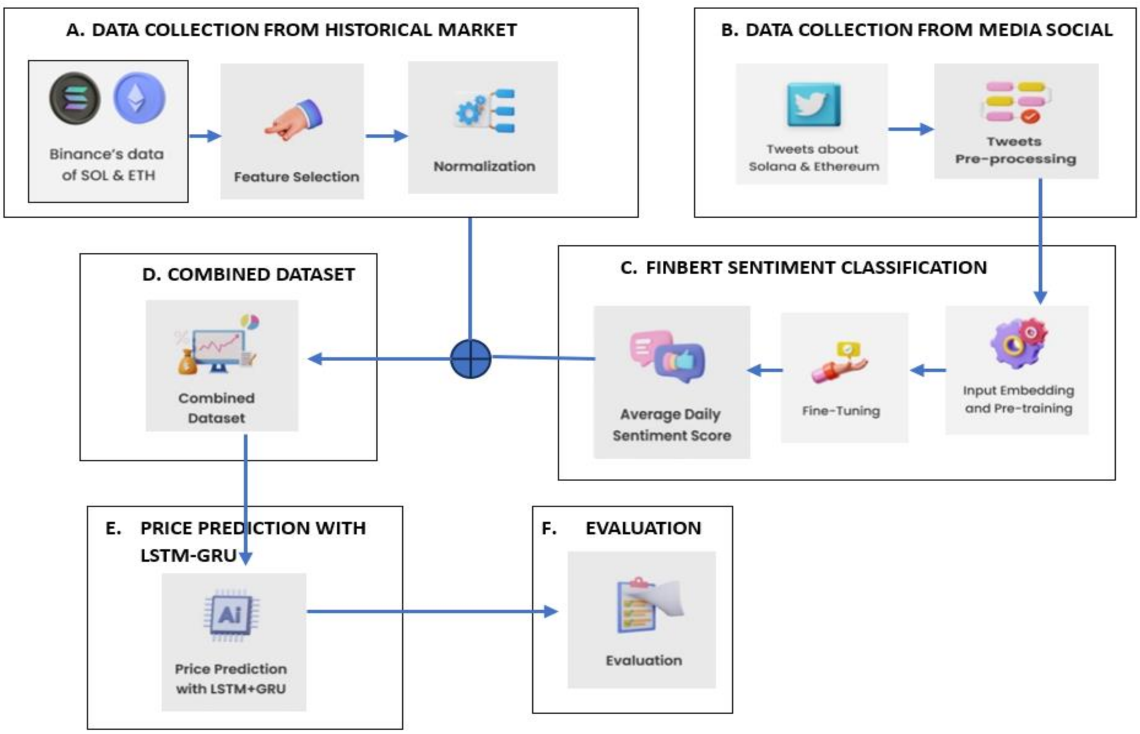

FIGURE 3. Proposed Method Steps

D. BERT

Bidirectional Encoder Representations from Transformers, or commonly abbreviated as BERT is a machine learning model that is engaged in the scope of Natural Language Processing (NLP) which also achieves state-of-the-art results in solving inputted and understand the context of each sentence. The training process for BERT to understand text data is Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). Before BERT is trained with MLM and NSP, all input texts must go through the input embeddings stage, the details of which can be seen in Figure 3. In MLM, 15% of words from each sentence are omitted and replaced by the [MASK] token. In this MLM, BERT is trained to be able to fill each [MASK] token with the correct word. The NSP process is a process where BERT is trained to understand the order of each word of each sentence. By understanding the position of each word in the sentence, BERT will be able to understand the meaning of each word according to the context of the sentence. Text input will go through the Sentence Embedding and Positional Embedding processes first to assist the NSP process. As depicted in Figure 3, each sentence and word are given a specific code to find out each sentence order and word order.

After BERT has understood the meaning and context of each input text, it will proceed to the part where the fine-tuning stage is needed. Fine tuning is the stage of modifying the output layer of the BERT architecture and adapting it to specific tasks such as sentiment classification, word classification, question and answer engine, and sentence marking, according to the dataset to be used.

E. FINBERT

FinBERT is a specialized language model based on the BERT (Bidirectional Encoder Representations from Transformers) architecture that is specifically designed for financial sentiment analysis and financial text classification tasks. It is trained on a large corpus of financial news articles, earnings call transcripts, and other financial text data [28].

FinBERT leverages the pretraining capabilities of BERT, which is a transformer-based model trained on a massive amount of general domain text data. However, FinBERT further fine-tunes the BERT model on financial text data to make it more adept at understanding the nuances of financial language and capturing financial sentiment.

The advantages of FinBERT lie in its ability to understand the unique vocabulary, jargon, and context of financial text. By pretraining on financial data and then fine-tuning on specific financial tasks, FinBERT can provide more accurate sentiment analysis and classification for financial text [33].

III. PROPESED METHOD

The proposed method is shown in Figure 3. The detailed steps are described in this sub section.

A. DATA COLLECTION FROM HISTORICAL MARKET

1. Binance's Data of SOL and ETH. Cryptocurrency historical market data of solana (SOL) and ethereum (ETH) are obtained from *www.cryptodatadownload.com* which contains data on the 28 cryptocurrencies market on Binance, one of the largest cryptocurrency trading sites in the world. Tweets and market data for Solana and Ethereum will be obtained separately and retrieved daily for a year, from January 1, 2021, to December 31, 2021.

2. Feature Selection. Some features can be obtained from www.cryptodatadownload.com. However, the feature selection is performed by choosing from all features. They are closing price, opening price, highest price, and lowest price.

3. Normalization. On the dataset that has been formed, the data normalization process or feature scaling will be carried out. The MinMaxScaling function provided by the Scikit-Learn library will be used. MinMaxScaling will homogenize numeric data that previously had values with different scales into values between 0 to 1. 0 represents the lowest value in the data feature and 1 represents the highest value, so that all data has a uniform value.

B. DATA COLLECTION FROM MEDIA SOCIAL

1. Tweet about Solana and Ehtereum. The data collected is in the form of text from Social network 's tweets with topics about Solana and Ethereum to be processed and used as sentiment data. The other data to be collected is historical cryptocurrency market data that contains information about price movements in daily timeframe. Tweets data is obtained directly from Social network by using a python library called SNScrape that will collect social network data according to the search query that entered by user. In this case, up to 50 tweets which have a minimum of 20 retweets will be collected per day. Also, only tweets that contain the words “ethereum”, “solana”, “eth”, or “sol” is included, and tweets with the word “giveaway" or “airdrop” will be filtered out to minimize outlier data.

2. Tweets Pre-processing. The tweet data that has been obtained must go through the pre-processing stage before sentiment extraction is carried out. This stage of data preprocessing is very important and can improve the accuracy of sentiment classification. This is because tweet data has attributes that cannot be considered features and have no influence in determining sentiment. Therefore, the data must be cleaned first before going to the next process. The preprocessing stage on social network data includes removing signs or symbols such as "#" and "@", removing the "RT" retweet sign, deleting newline marks, and removing all URLs or links.

C. FINBERT SENTIMENT CLASSIFICATION

1. Input Embedding and Pre-Training. Tweets data from social network that has been cleaned through the preprocessing stage will then be processed by one of the pre-trained models from BERT, namely FinBERT. By FinBERT, the text data that has been obtained will go through several stages. The first stage is to train text input data in a general scope (general corpus). With various text data from various Book Corpus and Wikipedia.

2. Fine Tuning. The second stage is to adapt text data by training it in a financial context with a dataset from Reuters TRC-2-financial containing 1.8 million financial news articles. After the data has been trained in the financial corpus, the data will be trained to perform the classification task. This classification training was conducted using the Financial

PhraseBank sentiment dataset which contains 4845 English sentences from various financial news [28].

3. Average Daily Sentiment Score. Data Finbert will get the sentiment score for each tweet that ranges from 0 to 1, close to 0 means sentiment is close to negative and close to 1 means sentiment is close to positive. Since the dataset that will be used is data for the daily period, the sentiment values of various tweets per day will be averaged so that daily sentiment score will be obtained which will later be combined with selected market data.

D. COMBINED DATASET

Combined dataset is got from historical market data combining with the average daily sentiment tweet. The historical market data selected for prediction features are closing prices, opening price, lowest price, and highest price, all in the daily time. **The closing price** of a cryptocurrency is the last recorded price at the end of a specific time period, typically a day. It is an important feature for price prediction because it encapsulates all the trading activity and market sentiment that occurred during that time frame. Traders and investors often use closing prices to identify trends, support and resistance levels, and overall market sentiment.

TABLE 1. Features of Combined Dataset

Feature	Description
Close	Nominal daily closing price
Open	Nominal daily opening price
High	Nominal daily higest price
Low	Nominal daily lowest price
Sentiment	Social network daily sentiment score

By analyzing the historical closing prices, predictive models can capture patterns and trends that might repeat in the future, helping to forecast potential price movements. The opening price of a cryptocurrency is the first recorded price at the beginning of a trading period, usually a day. It's significant because it sets the initial tone for trading activity and reflects the market sentiment at the start of the day. **The opening price** can be influenced by news, events, and sentiments from the previous day or during non-trading hours. Including the opening price as a feature in prediction models allows them to capture the immediate reaction of the market to new information, which can be crucial for short-term price movements. **The lowest price of a cryptocurrency** during a specific time period represents the point at which the market saw the least demand or the most selling pressure. This feature is valuable for prediction because it indicates the level at which buyers stepped in to support the price, potentially acting as a support level in the future. Analyzing the historical lowest prices can help models identify key support levels that traders often pay attention to. Predictive models can use this information to estimate potential price rebounds or reversals when the price approaches historical intraday lows. **The highest price** of a cryptocurrency during a certain time frame signifies the peak of buying interest or positive market sentiment during that period. This feature is crucial for prediction as it shows the level of demand and enthusiasm among traders. The historical highest prices can help models identify potential resistance levels, where the price might encounter selling pressure in the future. These resistance levels can guide predictions about potential price pullbacks or consolidation phases.

Daily sentiment data that has been obtained will be combined with data on daily closing price, daily opening price, daily highest price, and daily lowest price of certain cryptocurrencies. Therefore, this combined dataset consists of 5 features variables and dates as the index variable as shown in Table 1.

E. PRICE PREDICTION WITH LSTM-GRU

The dataset that has 5 variables will be used as vector form to be input to the prediction algorithm as time series data. The algorithm used to make predictions is an algorithm derived from the Recurrent Neural Network (RNN), which is a hybrid model of LSTM and GRU. Both algorithms are known for their ability to remember important information from previous data iterations. The algorithm derived from the RNN has also been proven to have good performance in making time series forecasting type predictions.

As can be seen in figure 4, the data will be trained by the LSTM and GRU separately with 30 neurons. The output of both will then be fed to the dropout layer to prevent overfitting. The result of the dropout layer via GRU will be followed by a dense layer. The results of the dropout layer via LSTM will be fed to the LSTM layer again with 50 neurons, then continued with a dense layer. The output from the dense layer via LSTM and GRU will then be concatenated and will return through the dense layer with ReLu as the activation function used. Then the prediction results using the LSTM-GRU hybrid algorithm are obtained. Hybrid LSTM-GRU and all other models involved in this study use the following parameters, as shown in Table 2. Windows period values 7 (seven) because the prediction is used for a week. Batch size refers to the number of training examples used in a single iteration of the training process. In practice, different batch is tried sizes such as 8, 16, 32, 64, so forth. In this study. a batch size of 16 is used.

TABLE 2. Training Parameters

Parameter	Value
Epoch	100
Optimizer	Adam
Loss Function	MSE
Window Period	7
Batch Size	16

F. EVALUATION

Prediction results using LSTM-GRU on historical data and sentiment values was compared with LSTM-GRU prediction results on historical data without using sentiment, LSTM with and without sentiment, and also GRU with and without sentiment. Non hybrid LSTM and GRU model each have 50 neurons and follow the same training parameters as described in Table 2. This prediction comparison will use several error metrics, namely Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error (RMSE) with the aim of measuring the prediction error compared to the actual data. In these three metrics, the smaller the error number, the more accurate the prediction results. The three error measurement methods were chosen because they are often used as evaluation methods for time series forecasting [34]. Previous research related to the comparison of algorithms for cryptocurrency price predictions also used these evaluation methods [21].

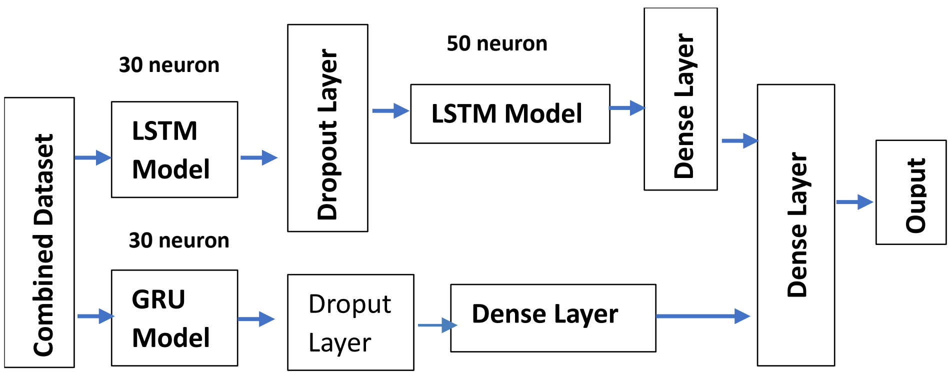

FIGURE 4. Hybrid LSTM-GRU Prediction Model Architecture

IV. EXPERIMENT RESULT

There are two parts experiment in this study. The first part is a comparison between several machine learning models without the sentiment score, and the second part is a comparison between the same several machine learning models, but with the sentiment score added to the historical dataset. Machine learning models that are used in the experiment are LSTM, GRU, and Hybrid LSTM-GRU. Both Ethereum and Solana Cryptocurrency are tested separately, but with the same treatment.

Table 3 and Table 4 show the experiment results of both Ethereum and Solana Cryptocurrency without sentiment. The testing is conducted five times for each algorithm. RNN algorithms such as LSTM and GRU usually produce inconsistent results due to their randomization element. To deal with it, each experiment is trained and tested five times, and the average value of the experiment will be used as a value for comparison. They show that GRU is better than LSTM, and hybrid GRU-LSTM is better than GRU. It can be seen that MAE, MAPE, and RMSE as the evaluation metrics to determine has the lowest error rate.

Table 5 and Table 6 show the experiment results of both Ethereum and Solana Cryptocurrency with sentiment. Without sentiment, GRU-LSTM results are also better than GRU and LSTM. To get the promising result of the experiment, and to make it easier to compare, results from five tests and train for Ethereum and Solana price has been averaged and summarized as shown in table 7 and table 8 above. Table 7 and table 8 shows MAE, MAPE, and RMSE of hybrid LSTM-GRU model with sentiment dataset is smallest for all test case. It shows that the proposed method is better than the hybrid LSTM-GRU which does not use daily sentiment score for the prediction. Not only for the hybrid LSTM-GRU, but all models that included the sentiment dataset have got a 0.5% to 1% MAPE improvement than models that trained without the sentiment dataset.

It's also shown that although hybrid LSTM-GRU has the best performance, GRU has better performance than LSTM for experiments using a sentiment dataset or not using a sentiment dataset. Another finding is that although including the sentiment dataset has better performance on LSTM, GRU, and hybrid LSTM-GRU, the performance is also highly dependent on the machine learning model used. This is shown in table 7 and table 8, where the performance for hybrid LSTM-GRU without sentiment dataset is better than LSTM or GRU with the sentiment. Line plot of the price prediction of Ethereum cryptocurrency which was made using Pyplot and Seaborn python library can be seen in figure 5. These plots are made as a visualization of the prediction results made by the LSTM-GRU hybrid algorithm using sentiment from social network . The blue and orange lines represent the results of training and testing, with a ratio of 80% of the data for training and 20% for testing and prediction. The predicted results are represented by a purple line. This study does not provide computation cost in this study because we do not calculate computation time. This research combines LSTM and GRU parallel which will definitely have a higher computation cost than LSTM and GRU individually.

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TABLE 3. Experiment Results (Ethereum Without Sentiment)

[Ethereum] Without Sentiment				
Test	Model	MAE	MAPE	RMSE
1	LSTM	326.49	7.72%	365.00
	GRU	257.86	6.13%	286.73
	LSTM-GRU	192.64	4.59%	214.63
2	LSTM	472.27	11.23%	497.86
	GRU	259.95	6.18%	292.48
	LSTM-GRU	202.28	4.82%	223.33
3	LSTM	298.61	7.07%	332.91
	GRU	243.48	5.80%	268.38
	LSTM-GRU	173.70	4.15%	199.70
4	LSTM	372.79	8.83%	405.91
	GRU	257.56	6.14%	281.82
	LSTM-GRU	224.26	5.34%	247.29
5	LSTM	342.54	8.12%	377.02
	GRU	282.15	6.72%	311.68
	LSTM-GRU	218.45	5.20%	241.48

TABLE 4. Experiment Results (Solana Without Sentiment)

[Solana] Without Sentiment				
Test	Model	MAE	MAPE	RMSE
1	LSTM	16.31	7.80%	18.95
	GRU	11.56	5.64%	13.27
	LSTM-GRU	9.95	4.81%	11.69
2	LSTM	14.14	6.78%	16.74
	GRU	14.10	6.88%	15.97
	LSTM-GRU	9.40	4.55%	11.18
3	LSTM	19.96	9.55%	22.83
	GRU	14.01	6.81%	15.86
	LSTM-GRU	9.86	4.79%	11.50
4	LSTM	15.00	7.18%	17.64
	GRU	13.77	6.69%	15.78
	LSTM-GRU	8.88	4.32%	10.61
5	LSTM	17.85	8.54%	20.63
	GRU	13.45	6.53%	15.51
	LSTM-GRU	9.83	4.75%	11.58

TABLE 5. Experiment Results (Ethereum With Sentiment)

[Ethereum] With Sentiment				
Test	**Model**	**MAE**	**MAPE**	**RMSE**
1	LSTM	290.87	6.87%	327.45
	GRU	266.54	6.33%	297.33
	LSTM-GRU	145.67	3.50%	173.69
2	LSTM	328.99	7.77%	364.94
	GRU	311.12	7.39%	338.41
	LSTM-GRU	162.08	3.88%	186.47
3	LSTM	245.14	5.78%	284.40
	GRU	200.88	4.78%	232.03
	LSTM-GRU	186.97	4.46%	211.70
4	LSTM	358.64	8.49%	393.40
	GRU	204.68	4.87%	236.26
	LSTM-GRU	186.53	4.45%	212.36
5	LSTM	300.66	7.10%	337.13
	GRU	171.32	4.09%	197.45
	LSTM-GRU	178.94	4.27%	203.28

TABLE 6. Experiment Results (Solana with Sentiment)

[Solana] With Sentiment				
Test	**Model**	**MAE**	**MAPE**	**RMSE**
1	LSTM	11.58	5.63%	14.16
	GRU	9.55	4.79%	11.77
	LSTM-GRU	8.28	4.04%	9.76
2	LSTM	13.61	6.53%	16.15
	GRU	15.34	7.47%	17.18
	LSTM-GRU	8.58	4.16%	10.09
3	LSTM	18.16	8.65%	21.05
	GRU	12.09	5.88%	14.15
	LSTM-GRU	8.49	4.13%	9.90
4	LSTM	20.89	9.99%	23.68
	GRU	9.06	4.49%	11.28
	LSTM-GRU	8.37	4.15%	10.13
5	LSTM	16.75	7.99%	19.53
	GRU	9.68	4.74%	11.62
	LSTM-GRU	8.35	4.16%	10.21

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TABLE 7. Experiment Results (Ethereum Average Error Value)

[Ethereum] Average Error Value				
Sentiment	**Model**	**MAE**	**MAPE**	**RMSE**
Not Included	LSTM	362.54	8.59%	395.74
	GRU	260.20	6.19%	288.22
	LSTM-GRU	202.27	4.82%	225.28
Included	LSTM	304.86	7.20%	341.46
	GRU	230.91	5.49%	260.30
	LSTM-GRU	172.04	4.11%	197.50

TABLE 8. Experiment Results (Solana Average Error Value)

[Solana] Average Error Value				
Sentiment	**Model**	**MAE**	**MAPE**	**RMSE**
Not Included	LSTM	16.65	7.97%	19.36
	GRU	13.38	6.51%	15.28
	LSTM-GRU	9.58	4.64%	11.31
Included	LSTM	16.20	7.76%	18.91
	GRU	11.14	5.47%	13.20
	LSTM-GRU	8.41	4.13%	10.02

FIGURE 5. Visualized Ethereum Price Prediction Result

V. CONCLUSION

This study proposed a new method to predict prices of two widely known cryptocurrencies, namely Ethereum and Solana. Our time-series prediction method utilizes the sentiment value contained in every tweet that discusses ethereum or solana. Sentiment values extracted using FinBERT will be grouped per day into daily sentiment values, which will then be combined with daily historical data. The combined dataset is then fed into the hybrid LSTM-GRU, one of the best models for conducting time-series forecasting which is sourced from previous research.

This research proves that the proposed method of adding sentiment score from Social network which extracted using FinBERT can improve prediction performance from only using commonly used time series prediction models such as LSTM, GRU, and even hybrids of the two. Social media is a platform where traders and investors express their opinions which also affect changes in asset prices. Most of the crypto price predictions that are usually done do not consider the sentiments of social media. The use of this sentiment data is expected to help researchers to improve the performance of the model and help traders and investors to maximize their profits.

This study has several limitations that can be used for future research. Sentiment taken is only sourced from social network , and there is difficulty in choosing relevant tweets, because there are some tweets that can be considered as outliers because they do not express sentiment well. For higher performance improvements in future research, sentiment for cryptocurrencies may be taken from several other sources such as google trends and forums as well as the crypto community, it is also possible that filtering for sentiment can be more focused and other sentiment extraction models can be used.

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